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SYSTEMATIC REVIEW

Do Sports Compression Garments Alter Measures of Peripheral Blood Flow? A Systematic Review with Meta-Analysis

Shane F. O'Riordan[1,2] · David J. Bishop[1] · Shona L. Halson[3] · James R. Broatch[1]

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Abstract

Background One of the proposed mechanisms underlying the benefits of sports compression garments may be alterations in peripheral blood flow.

Objective We aimed to determine if sports compression garments alter measures of peripheral blood flow at rest, as well as during, immediately after and in recovery from a physiological challenge (i.e. exercise or an orthostatic challenge).

Methods We conducted a systematic literature search of databases including Scopus, SPORTDiscus and PubMed/MEDLINE. The criteria for inclusion of studies were: (1) original papers in English and a peer-reviewed journal; (2) assessed effect of compression garments on a measure of peripheral blood flow at rest and/or before, during or after a physiological challenge; (3) participants were healthy and without cardiovascular or metabolic disorders; and (4) a study population including athletes and physically active or healthy participants. The PEDro scale was used to assess the methodological quality of the included studies. A random-effects meta-analysis model was used. Changes in blood flow were quantified by standardised mean difference (SMD) [±95% confidence interval (CI)].

Results Of the 899 articles identified, 22 studies were included for the meta-analysis. The results indicated sports compression garments improve overall peripheral blood flow (SMD = 0.32, 95% CI 0.13, 0.51, p = 0.001), venous blood flow (SMD = 0.37, 95% CI 0.14, 0.60, p = 0.002) and arterial blood flow (SMD = 0.30, 95% CI 0.01, 0.59, p = 0.04). At rest, sports compression garments did not improve peripheral blood flow (SMD = 0.18, 95% CI − 0.02, 0.39, p = 0.08). However, subgroup analyses revealed sports compression garments enhance venous (SMD = 0.31 95% CI 0.02, 0.60, p = 0.03), but not arterial (SMD = 0.12, 95% CI − 0.16, 0.40, p = 0.16), blood flow. During a physiological challenge, peripheral blood flow was improved (SMD = 0.44, 95% CI 0.19, 0.69, p = 0.0007), with subgroup analyses revealing sports compression garments enhance venous (SMD = 0.48, 95% CI 0.11, 0.85, p = 0.01) and arterial blood flow (SMD = 0.44, 95% CI 0.03, 0.86, p = 0.04). At immediately after a physiological challenge, there were no changes in peripheral blood flow (SMD = − 0.04, 95% CI − 0.43, 0.34, p = 0.82) or subgroup analyses of venous (SMD = − 0.41, 95% CI − 1.32, 0.47, p = 0.35) and arterial (SMD = 0.12, 95% CI − 0.26, 0.51, p = 0.53) blood flow. In recovery, sports compression garments did not improve peripheral blood flow (SMD = 0.25, 95% CI − 0.45, 0.95, p = 0.49). The subgroup analyses showed enhanced venous (SMD = 0.67, 95% CI 0.17, 1.17, p = 0.009), but not arterial blood flow (SMD = 0.02, 95% CI − 1.06, 1.09, p = 0.98).

Conclusions Use of sports compression garments enhances venous blood flow at rest, during and in recovery from, but not immediately after, a physiological challenge. Compression-induced changes in arterial blood flow were only evident during a physiological challenge.

✉ Shane F. O'Riordan
shane.oriordan@vu.edu.au

1 Institute for Health and Sport (iHeS), Victoria University, Melbourne, VIC, Australia

2 Department of Physiology, Australian Institute of Sport, Canberra, ACT, Australia

3 School of Behavioural and Health Sciences, Australian Catholic University, McAuley at Banyo, Brisbane, QLD, Australia

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Key Points

The results of this systematic review and meta-analysis suggest that sports compression garments enhance peripheral blood flow, particularly venous blood flow.

Compression-induced improvements in venous blood flow were evident at rest, during and in recovery from a physiological challenge, but not immediately after a physiological challenge timepoint.

Studies involving sports compression garments present moderate methodological quality because of the lack of blinding of participants and researchers.

1 Introduction

Sports compression garments (SCG) are used in athletic settings to enhance exercise performance and recovery [1]. These garments apply pressure to the area they are covering, typically increasing in pressure from the distal to proximal portions of the limb (i.e. graduated compression) [2]. The use of SCG during exercise has been reported to improve numerous performance metrics, including jump height [3–6], cycling power [7–10], repeated sprint ability [8, 11] and running performance [12–14]. When worn during the post-exercise recovery period, compression garments are reported to reduce ratings of muscle soreness [15–20], decrements in subsequent exercise performance [17, 20–22] and muscle swelling [16, 20, 23], as well as enhance the clearance of muscle damage markers [16, 23–25]. Collectively, it has been hypothesised that the ergogenic effects of SCG during and following exercise are largely associated with compression-induced increases in peripheral blood flow (i.e. venous and arterial blood flow) [26, 27].

Compression-induced increases in venous blood flow [28–30] occur through mechanical distortion of the underlying vessels by decreasing vein diameter and improving valve competence [31]. As a result, blood is diverted from superficial veins, through perforator veins, and into deep veins leading to an increase in deep venous velocity, reduced venous pooling, and improved venous return [32, 33]. In support of this mechanism, graduated compression stockings increased resting popliteal venous flow by ~30%, coinciding with a decrease in venous cross-sectional area [28]. However, compression alters arterial blood flow via a different mechanism. In essence, the application of compression increases arterial vessel size (i.e. vasodilation) [34]. This reflex vasodilation, termed a myogenic response, is proposed to occur owing to a reduction in arteriolar transmural pressure [35]. Subsequently, decreasing arterial flow resistance leads to increased arterial blood flow, thus improving measures of blood supply to the muscle and the capillary networks (e.g. muscle blood flow and microvascular perfusion) [35].

The external pressure applied to the limb by SCG has been shown to increase venous and arterial blood flow measures in some [8, 28, 29, 35–38], but not all studies [39–41]. For instance, we reported sports compression tights to increase muscle blood flow, measured using near-infrared spectroscopy (NIRS) with venous occlusions, by ~18% during repeated cycling exercise [8]. Conversely, using the same exercise protocol [8], we recently observed a compression-induced decrease in muscle microvascular perfusion, measured using contrast-enhanced ultrasound (CEU), despite increased femoral artery blood flow [34]. Additionally, while many studies report compression to enhance venous [28, 29, 36] and arterial [8, 35, 37, 38] blood flow measures, solid conclusions have yet to be established, with some reports showing SCG do not alter venous or arterial blood flow [39–41]. Our recent observations [8, 34] highlight that variations in blood flow measurement techniques may help explain these inconsistent findings. Additionally, the contrasting reports are likely due to the high level of heterogeneity in SCG details and the intensity of physiological challenges used in compression research.

Characteristics of SCG, including the degree of pressure applied and body area covered (i.e. upper body, calf area, entire lower-limb), determine the effectiveness of the garment in altering blood flow [28, 42]. In support of this concept, differences in venous blood flow were observed between light (14.0 mmHg), mild (21.2 mmHg), moderate (32.1 mmHg) and high (46.5 mmHg) degrees of pressure at the ankle in graduated compression stockings [28]. Compression shorts exerting high pressure (37 mmHg) have also been reported to reduce muscle blood flow [43]. Additionally, as blood flow increases to meet the metabolic demand of the active muscle [44, 45], the intensity of a physiological challenge (i.e. exercise) may influence the efficacy of SCG to alter peripheral blood flow. For example, compression-induced increases in muscle blood flow have been reported during repeated sprint exercise [8], but not during prolonged trail running [39]. The differences in SCG type (tights for repeated sprint exercise, socks for prolonged trail running) may also help explain these contrasting findings.

To date, most systematic reviews and meta-analyses investigating the ergogenic effects of SCG have focused on exercise performance and/or recovery outcomes [26, 27, 46–49]. Although the benefits of SCG during and following exercise are likely related to changes in blood flow, only one systematic review has focused on haemodynamic outcomes [50]. This review concluded that wearing compression garments positively alters central haemodynamic

responses (i.e. heart rate and stroke volume) at rest or after a physiological challenge. It was also noted that compression garments increase stroke volume and decrease heart rate, with responses more pronounced after a physiological challenge. However, compression did not affect other central haemodynamic measures, including cardiac output, blood pressure or systemic vascular resistance. Alterations in central haemodynamics with SCG may result in, and/or be a result of, increases in peripheral blood flow. For example, the augmented activity of the skeletal muscle pump with compression increases venous flow velocity and enhances venous return [28]. In turn, end-diastolic filling and stroke volume are increased, thus reducing heart rate [8, 51]. However, collective data regarding the influence of SCG on peripheral blood flow measures are yet to be synthesised. Thus, a systematic review and meta-analysis of the available literature is warranted to determine the effect of SCG on measures of peripheral blood flow and to provide a more comprehensive understanding of the primary mechanism underlying the ergogenic effects of SCG.

The aim of this systematic review with meta-analysis was to systematically evaluate the existing literature that has investigated the effects of SCG on peripheral measures of blood flow. Additionally, this review and analysis aimed to (a) identify any differences in peripheral blood flow with SCG at rest and during, immediately after or in recovery from a physiological challenge and (b) highlight methodological variations in compression research, including SCG details (i.e. type and degree of pressure), physiological challenges and the techniques used to assess blood flow.

2 Methods

This systematic review was conducted according to Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines [52].

2.1 Literature Search

A literature search was performed using the following online databases: Scopus, SPORTDiscus and PubMed/MEDLINE. The following syntax of keywords used to find relevant papers was: ("compression garments" OR "compression clothing" OR "sports compression garments" OR "graduated compression" OR "compression tights" OR "compression socks" OR "compression stockings" OR "compression shorts" OR "compression sleeves") AND ("blood flow" OR "haemodynamic" OR "cardiovascular" OR "oxygenation" OR "venous" OR "arterial" OR "perfusion") AND ("exercise" OR "recovery" OR "rest"). Advanced search techniques from the search engines of "related words" were also included. The search of databases ended in January 2022. The reference list of obtained articles and reviews was examined to identify any further relevant studies to supplement the database searches. The study selection was independently performed by two authors (SOR and JB) to prevent selection bias, and any disagreement resolved by third-party adjudication (SH).

2.2 Inclusion and Exclusion Criteria

Studies were included in the review on the basis of the following criteria: (1) original papers in English and in a peer-reviewed journal; (2) assessed the effects of compression garments on a measure of peripheral blood flow at rest (with garments used in a physiological challenge or described as SCG only) and/or prior to, during or following a physiological challenge (i.e. exercise or an orthostatic challenge); (3) participants were healthy and did not have any cardiovascular or metabolic disorders; and (4) a study population including physically active (e.g. athletes, recreationally active) or healthy participants. Studies were excluded if: (1) changes in peripheral blood flow were not assessed while wearing compression garments; (2) compression garments were used in conjunction with other treatments (e.g. nutrition, cold water immersion, stretching); (3) no control condition was included; and (4) insufficient data were available to conduct the meta-analysis after contact with the authors.

2.3 Data Extraction

The following data were extracted from the selected studies for SCG and control groups that met the inclusion criteria: (1) participant characteristics of samples size, sex, age and physical activity level as described in the study (healthy, recreationally active or participated in sport); (2) compression garment details including type (e.g. tights, shorts, socks), pressure (measured or manufacturer reported values in mmHg) and when compression garments were worn (rest, during, immediately after and/or in recovery from a physiological challenge); (3) outcome variable to assess compression effects on peripheral blood flow at any of the following timepoints: rest (before a physiological challenge), during (during a physiological challenge), immediately after (cessation of a physiological challenge or described as post) and recovery (timepoint after a physiological challenge or described as recovery); and (4) physiological challenge details (mode, duration, intensity and distance). Peripheral measures of blood flow were identified as any measurement representative of blood flow (e.g. flow, volume, velocity, perfusion) through blood vessels within the heart or part of the systemic circulatory system. For studies that included multiple measures of either venous or arterial blood flow, the data of the measurement that best represents vessel blood flow or that is influenced by other measures were extracted

(e.g., arterial blood flow is included instead of arterial time averaged mean velocity). Also, for studies that investigated different garment pressures (e.g., regular sized compared to undersized garments), the data from the compression garment that represent typical pressures (i.e., not higher than normal) were extracted. When required, the Web Plot Digitiser software (Version 4.4; TX, USA; Ankit Rohatgi, 2020) was used to extract data from figures. All data were tabulated in an Excel spreadsheet (Microsoft Corporation, Redmond, WA, USA) independently by two authors (SOR and JB). Files were cross-checked between the authors, with discrepancies resolved through discussion and agreement, or third-party adjudication (SH).

2.4 Risk of Bias Assessment and Methodological Quality

For the assessment of the methodology quality of studies, two authors (SOR and JB) used the 11-point Physiotherapy Evidence Database (PEDro) scale [53]. Consultation with a third author (SH) was used to resolve disagreements. The maximal score a study could achieve was 10 points. The first statement was excluded from the overall score as it concerns whether the participant eligibility criteria were specified and does not represent the methodological quality of the study (assessed with items 2–11). Studies were then classified as having excellent (9–10), good (6–8), moderate (4–5) or poor (< 4) methodological quality [54]. The PEDro scale is a valid and reliable method to assess the quality of randomised controlled trials [53, 55], and has been used previously in compression garment systematic reviews [27, 48, 50, 56].

2.5 Statistical Analysis

The average value of the means and average standard deviation for each selected outcome measure were calculated for both SCG and control groups within individual studies. The average standard deviation was calculated using guidelines highlighted in the *Cochrane Handbook for Systematic Reviews* [57]. In each study, the mean change and standard deviation [58] from rest values to during, immediately post and recovery were determined for venous and arterial blood flow measures in the SCG and control groups. Studies that investigated peripheral blood flow responses at numerous time points during (i.e., set 1, set 2) or in recovery, the peak change value and the final time point were used for during and recovery analysis, respectively. These values, in addition to rest values, were used in RevMan 5 (Review Manager, Version 5.4; Cochrane Collaboration) with a random-effects model to calculate the standardised mean difference (SMD) and 95% confidence intervals (CIs) between groups (SCG vs control) during, immediately post and in recovery. A random-effects model was chosen to calculate the average distribution of effects that can be expected as the effects of peripheral blood flow may vary according to participant training status, physiological challenge, measurement techniques and SCG characteristics. At each timepoint, a subgroup analysis (venous or arterial blood flow) was included to quantify the SCG effect on venous or arterial measures of peripheral blood flow. Standardised mean differences were defined as trivial (0.00–0.19), small (0.20–0.49), medium (0.50–0.79) or large (≥ 0.80) [59]. Heterogeneity was assessed using Cochran's Q (a chi-square test of residual heterogeneity), I^2 statistic (proportion of the residual heterogeneity on unaccounted variability) and Tau^2 (standard deviation from the study-estimate random effect). Heterogeneity among studies was evaluated by the I^2 statistic as follows: low heterogeneity was assumed when $I^2 < 25\%$; moderate heterogeneity when $I^2 < 75\%$ and $> 25\%$; and high heterogeneity when $I^2 \geq 75\%$ [60]. A significance level of $p < 0.05$ was applied.

3 Results

3.1 Included Studies

A total of 895 articles were identified through database searches, with a further four articles identified through a manual search of reference lists. Of the 899 articles identified, 288 were duplicates, and 548 were excluded after review of their title and abstract. A total of 63 full-text articles were assessed for eligibility (Fig. 1).

3.2 Data Extraction and Risk of Bias Assessment

Of the 63 relevant articles, 41 were excluded because of an incorrect intervention (e.g. compression garments used in conjunction with other interventions), not in English, wrong outcomes (e.g. not a measure of peripheral blood flow), missing data, incorrect cohort (e.g. patients with blood flow pathologies) or wrong study design (e.g. no control condition) (Fig. 1). Twenty-two articles that met the inclusion criteria had data extracted for inclusion in the meta-analysis (Table 1). Across the 22 studies, 23 different measures of peripheral blood flow were identified. These included arterial blood flow (ABF) [34, 37, 61, 62], arterial reserve (AR) [61], arterial time-averaged mean velocity (A-TAMv) [34, 37], deep muscle blood flow (mBF_{deep}) [43], ejection fraction (EF) [63, 64], end diastolic velocity (EDv) [37], middle cerebral artery blood flow velocity (MCAv) [65–67], microvascular blood flow (MVBF) [34, 68], microvascular blood velocity (MVv) [34], microvascular blood volume (MVV) [34], muscle blood flow (mBF) [8, 35, 40, 42], peak systolic velocity (PSv) [37], mean artery blood velocity (PBv_{mean}) [41], mean artery flow rate (PBF_{mean}) [41], vein

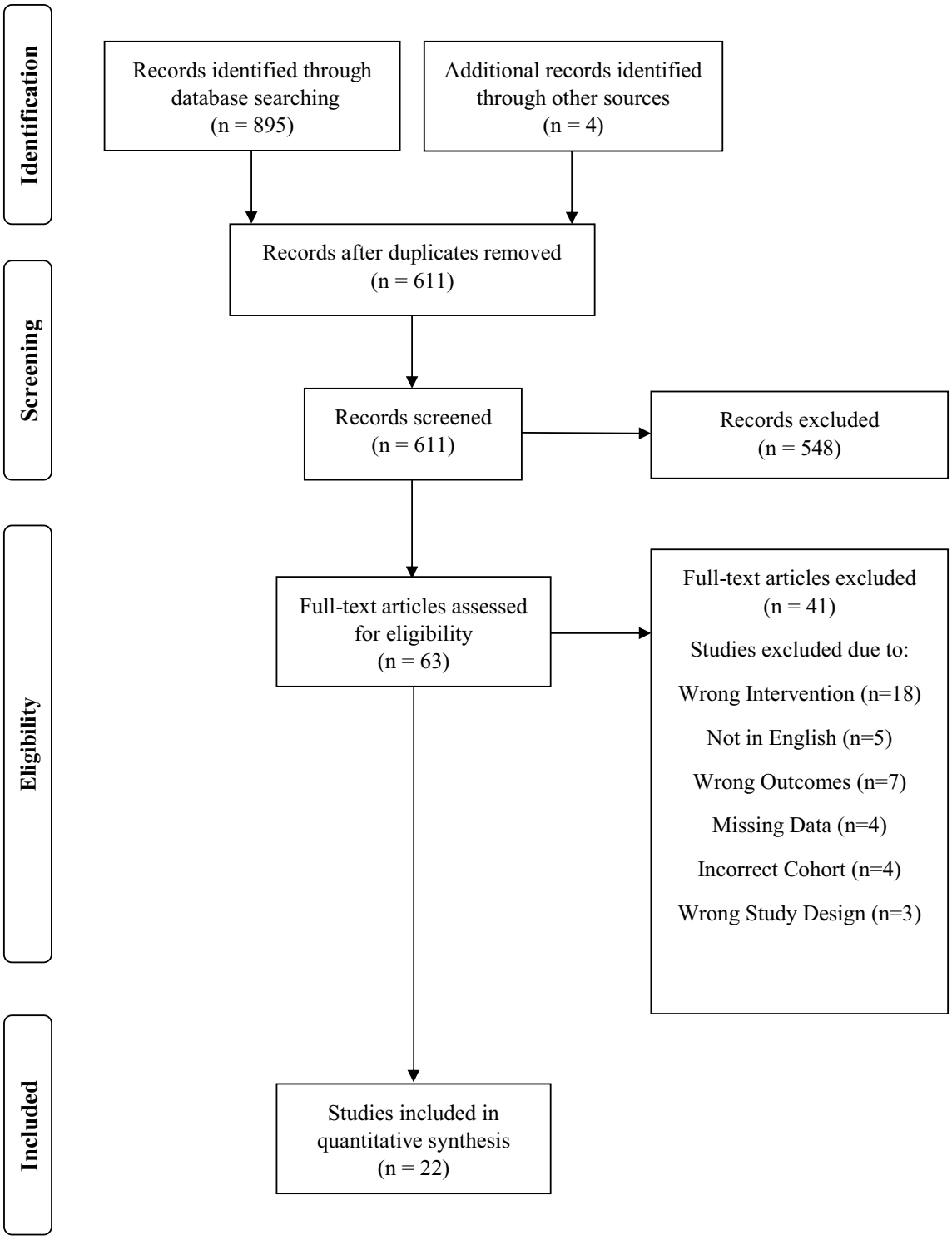

Fig. 1 Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) flow diagram of the literature screening process

blood flow (VBF) [42, 69], residual volume fraction (RVF) [63], resistance index (RI) [37], superficial muscle blood flow ($mBF_{superficial}$) [43], total haemoglobin (tHb) [14, 34, 36, 41, 70, 71], vein time-average mean velocity (V-TAMv) [42], venous emptying rate (VER) [62], venous filling index (VFI) [63] and venous reserve volume (VRV) [62]. A total of six different styles of compression garments were used to assess changes in peripheral blood flow among the 22 studies. These styles were tights (covering from above ankle to waist) [8, 34, 36, 37, 42, 62, 66, 67, 70, 72], socks (covering

Table 1 Characteristics and summary of studies included in the meta-analysis

Study	Cohort, sample size, sex, age (years)	Garment type	Pressure (mmHg)	Physiological challenge (timing and duration of compression)	Peripheral blood flow variable	Effect (timepoint)
Bochmann et al. [35]	Healthy 5M 25.0 ± 3.0	Arm sleeve	Forearm: 16.0 ± 5.0[a]	70-min rhythmic hand exercise (rest, during)	mBF	↑ (pre), ↑ (during)
Book et al. [41]	Healthy 5F, 7M 27.3 ± 6.4	Socks	Ankle: 30.8 ± 5.8[a] Knee: 20.2 ± 5.7[a]	5-min plantar flexion (rest, during, recovery: 5 min)	tHb* A-TAMV ABF*	↔ (pre, recovery), ↑ (during) ↔ (pre, during, recovery) ↔ (pre, during, recovery)
Born et al. [72]	Ice speed skaters 6F, 4M 23.0 ± 7.0	Tights	Thigh: 20.3 ± 2.3[a] Calf: 24.4 ± 3.1[a]	3000-m ice speed skating race (during)	tHb	↔ (during)
Bringard et al. [36]	Sportsmen 12M 26.5 ± 2.6	Tights	*Elastic tight* Calf: 5[a] *Compression tight** Calf: 24.1[a]	Supine rest (during)	tHb	↑ *Compression tight only*
Broatch et al. [8]	Recreationally active 11F, 9M 26.5 ± 6.3	Tights	Mid-thigh: 11.7 ± 2.3[a] Calf: 26.4 ± 6.4[a] Ankle: 21.5 ± 8.2[a]	Repeated sprint cycling (rest, during)	mBF	↔ (pre) ↑ (during)
Broatch et al. [34]	Recreationally active 2F, 8M 27.4 ± 6.3	Tights	Mid-thigh: 11.3 ± 3.1[a] Calf: 20.6 ± 3.6[a]	Repeated sprint cycling (rest, during, recovery: 60 min)	tHb* MVBF* MVV MVv A-TAMv ABF*	↔ (pre, recovery), ↓ (post) ↔ (pre), ↓ (post, recovery) ↔ (pre, post, recovery) ↔ (pre), ↓ (post, recovery) ↔ (pre, during, recovery) ↔ (pre, recovery), ↑ (during)
Castilho-Junior et al. [63]	Amateur runners 6F, 4M 40.3 ± 5.0	Socks	Below knee: 20[b] Ankle: 30[b]	10-km run (during)	VFI EF* RVF	↑ (post) ↔ (post) ↑ (post)
Dascombe et al. [70]	Middle-distance runners 11 M 28.4 ± 10.0	Tights	*Regular garment* Thigh: 13.7 ± 2.3[a] Calf: 19.2 ± 3.2[a] *Undersize garment* Thigh: 15.9 ± 2.6[a] Calf: 21.7 ± 4.3[a]	Time to exhaustion running at 90% VO_{2max} (during)	tHb	↑ (during)
Dascombe et al. [71]	Elite kayakers 2F, 5M 25.0 ± 4.2 (F) 21.8 ± 2.8 (M)	Upper body	n.r.	Kayak ergometer: six step incremental test plus 4-min performance test (during)	tHb	↔ (during)
Dorey et al. [65]	Healthy 4F, 6M 23.0 ± 2.0	Socks	Lower leg: 30–40[b]	OC (15-min 60° HUT), 60-min cycling at 60% peak workload, and OC (rest, during)	MCAv	↔ (pre, during) ↑ (post)
Kerherve et al. [40]	Trail runners 14M 21.7 ± 3.0	Socks	Calf: 23.0 ± 2.0[a]	24-km trail run (rest, during)	mBF	↑ (pre) ↔ (during)

Table 1 (continued)

Study	Cohort, sample size, sex, age (years)	Garment type	Pressure (mmHg)	Physiological challenge (timing and duration of compression)	Peripheral blood flow variable	Effect (timepoint)
Mann et al. [61]	Amateur runners 12F, 18M 23.0 (F), 24.0 (M)	Socks	n.r.	Incremental running test (during)	ABF* AR	↑ (post) ↑ (post)
Menetrier et al. [37]	Endurance trained 15M 22.5 ± 0.7	Tights	Thigh: 14[b] Calf: 27[b] Ankle: 15[b]	9 × 5-min cycling intervals (recovery: 12 min)	ABF* A-TAMv EDv* PSv RI	↑ (recovery) ↑ (recovery) ↔ (recovery) ↑ (recovery) ↑ (recovery)
Morrison et al. [66]	Healthy 15M 27.0 ± 4.0	Tights	10–15[a]	OC: supine to stand test (rest, during)	MCAv	↔ (pre, during)
Oficial-Casado et al. [69]	Runners 10M 35.0 ± 5.0	Socks	Medium* or high compression n.r.	30 min at 75% of maximal aerobic speed (rest, during)	Pop. VBF	↔ (pre, post)
O'Riordan et al. [42]	Basketball players 22M 17.2 ± 0.9	Socks*	Calf: 29.0 ± 2.5[a] Mid-shank: 25.8 ± 4.1[a] Ankle: 25.9 ± 4.5[a]	Supine rest (during)	Pop. V-TAMv Pop. VBF Fem. V-TAMv Fem. VBF* Calf mBF* Thigh mBF*	↑ Socks, tights ↔ shorts ↑ Tights ↔ socks, shorts
		Shorts*	Upper thigh: 11.6 ± 1.8[a] Mid-thigh: 13.0 ± 1.2[a] Lower thigh: 13.6 ± 1.6[a]			↑ Socks, shorts, tights ↑ Shorts, tights ↔ socks ↑ Socks, tights ↔ shorts ↑ Shorts, tights ↔ socks
		Tights*	Upper thigh: 12.2 ± 2.3[a] Mid-thigh: 13.0 ± 2.0[a] Lower thigh: 13.6 ± 2.4[a] Calf: 17.2 ± 3.9[a] Mid-shank: 17.2 ± 3.7[a] Ankle:15.1 ± 3.8[a]			
Partsch et al. [64]	*Exp 1:* Soccer players 12M n.r *Exp 2:* Tennis players 6M n.r.	Socks	Calf: 23[a]	Walking test (post)	EF	Exp 1: ↔ (post) Exp 2: ↔ (post)
Riexinger et al. [68]	Healthy 7F, 9M 24.1	Socks	Calf: 21–22[b]	Plyometrics and resistance: 5 × 10 drop jumps, 5 × 50 heel raises (rest, during, recovery: 6 h)	Calf MVBF* Tibialis MVBF	↔ (pre, post, recovery)

Table 1 (continued)

Study	Cohort, sample size, sex, age (years)	Garment type	Pressure (mmHg)	Physiological challenge (timing and duration of compression)	Peripheral blood flow variable	Effect (timepoint)
Sear et al. [14]	Team sport athletes 8M 20.6 ± 1.2	Full body	Medial forearm: 5.8 ± 1.0[a] Medial bicep: 7.3 ± 2.5[a] Medial chest: 5.3 ± 0.5[a] Oblique: 5.9 ± 0.8[a] Glute: 9.2 ± 1.6[a] Mid-thigh: 13.1 ± 1.7[a] Calf: 15.1 ± 2.0[a] Ankle: 17.8 ± 2.2[a]	45-min prolonged high-intensity intermittent exercise (during)	tHb	↔ (during)
Smale et al. [67]	Trained cyclists 15M 28.1 ± 6.3	Tights	Thigh: 15.4 ± 4.5[a] Knee: 20.3 ± 6.6[a] Ankle: 21.8 ± 6.6[a]	4 × 8-min incremental cycling (rest, during)	MCAv	↔ (pre, during)
Sperlich et al. [43]	Physically active 6M 22.0 ± 2.0	Shorts	Thigh: 36.7 ± 4.1[a]	Incremental cycling test (rest: 20 min, recovery: 35 min)	mBF	↔ (pre), ↔ (recovery) ↔ (pre), ↓ (recovery)
Venckunas et al. [62]	Physically active 13F 25.1 ± 4.2	Tights	Thigh: ~ 17[a] Calf: ~ 19[a]	4-km run in 30-min plus 400-m sprint (rest, during, recovery: 30 min)	ABF* VRV VER*	↓ (pre), ↔ (recovery) ↔ (pre, recovery) ↔ (pre), ↑ (recovery)

A-TAMv arterial time-averaged mean velocity, *ABF* arterial blood flow, *Ankle* medial malleolus of the ankle, *AR* arterial reserve, *Calf* maximal calf girth, *EDv* end diastolic velocity, *EF* ejection fraction, *F* female, *Fem.* femoral, *h* hours, *HUT* head up tilt test, *M* male, *mBF* muscle blood flow, *MCAv* middle cerebral artery blood flow velocity, *min* minute, *MVBF* microvascular blood flow, *MVv* microvascular blood velocity, *MVV* microvascular blood volume, *n.r.* not reported, *OC* orthostatic challenge, *Pop.* popliteal, *PSv* peak systolic velocity, *RI* resistance index, *RVF* residual volume fraction, *tHb* total haemoglobin, *Thigh* maximal thigh girth, *V-TAMv* vein time-averaged mean velocity, *VBF* vein blood flow, *VER* venous emptying rate, *VFI* venous filling index, *VRV* venous reserve volume, VO_{2max} maximal oxygen uptake, ↔ no significant effect from compression ($p < 0.05$), ↑ a significant positive effect from compression ($p < 0.05$), ↓ a significant negative effect from compression ($p < 0.05$)

*Peripheral blood flow measure or sports compression garment included in meta-analysis

[a]Pressure measured directly

[b]Manufacturer reported pressure

foot to below the knee) [40–42, 61, 63–65, 68, 69], shorts (covering from knee to waist) [42, 43], upper-body (covering torso plus entire arm to the wrist) [71], an arm sleeve (covering from wrist to elbow) [35] and a whole-body garment (combination of tights and upper-body garments) [14]. For the 22 included studies, the mean (± standard deviation) PEDro score was 5.8 ± 0.4 (range 5–6) (Table 2).

3.3 Participant Characteristics

The 22 studies included in this meta-analysis comprised a total of 289 participants (68 were female and 221 were male). One study included female participants only [62], 12 studies included male participants only [14, 35–37, 40, 42, 43, 64, 66, 67, 69, 70], and nine studies included both sexes [8, 34, 41, 61, 63, 65, 68, 71, 72]. Participant cohorts were healthy [35, 41, 65, 66, 68], physically active [8, 34, 43, 62] or participated in sport [14, 36, 37, 40, 42, 61, 63, 64, 67, 69–72]. The number of participants per study ranged from 5 to 30 (median: 12 participants), and the age of participants ranged from 17 to 40 years (median: 25 years).

3.4 Influence of SCG on Peripheral Blood Flow

Meta-analytic outcomes for the pooled average and standard deviations of the selected peripheral blood flow measures in each study are presented in Fig. 2 to provide an overall effect size value for the influence of SCG on each measure of peripheral blood flow. Peripheral blood flow measures from the 22 included studies showed a small positive effect for SCG compared with the control (SMD = 0.32, 95% CI 0.13, 0.51, $z = 3.28$, $p = 0.001$) (Fig. 2). Moderate heterogeneity amongst studies was observed (Cochran's $Q = 47.28$, $I^2 = 41\%$, $\text{Tau}^2 = 0.11$, $p = 0.01$). A total of 9 different measures of blood flow were assessed, including 5 venous (tHb [14, 34, 36, 41, 70–72, EF [63, 64], VBF [42, 69 VER

Table 2 Quality assessment of included studies based on the PEDro scale

PEDro item number													
Study	Score	Methodological quality	1*	2	3	4	5	6	7	8	9	10	11
Bochmann et al. [35]	5	Fair	✓			✓				✓	✓	✓	✓
Book et al. [41]	6	Good	✓	✓		✓				✓	✓	✓	✓
Born et al. [72]	6	Good	✓	✓		✓				✓	✓	✓	✓
Bringard et al. [36]	6	Good	✓	✓		✓				✓	✓	✓	✓
Broatch et al. [8]	6	Good		✓		✓				✓	✓	✓	✓
Broatch et al. [34]	6	Good	✓	✓		✓				✓	✓	✓	✓
Castilho-Junior et al. [63]	5	Fair	✓			✓				✓	✓	✓	✓
Dascombe et al. [70]	6	Good	✓	✓		✓				✓	✓	✓	✓
Dascombe et al. [71]	6	Good	✓	✓		✓				✓	✓	✓	✓
Dorey et al. [65]	6	Good		✓		✓				✓	✓	✓	✓
Kerhervé et al. [40]	6	Good	✓	✓		✓				✓	✓	✓	✓
Mann et al. [61]	6	Good	✓	✓		✓				✓	✓	✓	✓
Ménétrier et al. [37]	6	Good	✓	✓		✓				✓	✓	✓	✓
Morrison et al. [66]	6	Good	✓	✓		✓				✓	✓	✓	✓
Oficial-Casado et al. [69]	6	Good	✓	✓		✓				✓	✓	✓	✓
O'Riordan et al. [42]	6	Good	✓	✓		✓				✓	✓	✓	✓
Partsch et al. [64]	5	Fair	✓			✓				✓	✓	✓	✓
Riexinger et al. [68]	6	Good	✓	✓		✓				✓	✓	✓	✓
Sear et al. [14]	6	Good	✓	✓		✓				✓	✓	✓	✓
Smale et al. [67]	6	Good	✓	✓		✓				✓	✓	✓	✓
Sperlich et al. [43]	6	Good		✓		✓				✓	✓	✓	✓
Venckunas et al. [62]	6	Good	✓	✓		✓				✓	✓	✓	✓

1 = eligibility criteria specified, 2 = random allocation of the participants; 3 = blind allocation; 4 = similar baseline characteristics between groups; 5 = blinding of participants; 6 = blinding of the researchers who administered the intervention participants; 7 = blinding of the researchers who measured outcome; 8 = adequate follow-up; 9 = intention to treat; 10 = results of between-group statistical comparisons are reported for at least one key outcome; 11 = measures of variability for at least one key outcome, *not included in the methodological score

[62]) and 4 arterial (mBF [8, 35, 40, 42, 43], ABF [34, 37, 41, 61, 62], MVBF [34, 68], MCAv [34, 68]). For sub-group analysis, SCG had a small positive effect on venous (SMD = 0.37, 95% CI 0.14, 0.61, $z = 3.16$, $p = 0.002$) and arterial (SMD = 0.30, 95% CI 0.01, 0.59, $z = 2.00$, $p = 0.04$) measures of blood flow. Low heterogeneity amongst studies assessing venous blood flow was observed (Cochran's $Q = 12.43$, $I^2 = 3\%$, $\text{Tau}^2 = 0.01$, $p = 0.41$), whereas moderate heterogeneity was observed for studies assessing arterial measures of blood flow (Cochran's $Q = 34.57$, $I^2 = 57\%$, $\text{Tau}^2 = 0.19$, $p = 0.003$).

A total of 14 studies were included for the analysis of peripheral blood flow measures at rest (Fig. 3). There was a non-significant trivial difference for SCG compared to controls at rest (SMD = 0.18, 95% CI − 0.02, 0.39, $z = 1.73$, $p = 0.08$). Moderate heterogeneity amongst studies was observed (Cochran's $Q = 35.70$, $I^2 = 41\%$, $\text{Tau}^2 = 0.10$, $p = 0.02$). In this analysis, a total of 7 different measures of blood flow were assessed, including three venous (tHb [36, 41], VBF [42, 69], VER [62]) and four arterial (mBF [8, 35, 40, 42, 43] MCAv [65–67], MVBF [34, 68], ABF [34, 62, 41]). Sub-group analysis revealed contrasting effects of SCG on venous and arterial measures. There was a small positive effect evident for SCG compared with controls for measures of venous blood flow (SMD = 0.31, 95% CI 0.02, 0.60, $z = 2.12$, $p = 0.03$). In contrast, there was a non-significant trivial difference in arterial blood flow between SCG and control groups (SMD = 0.12, 95% CI − 0.16, 0.40, $z = 0.83$, $p = 0.41$). Low heterogeneity was observed for studies assessing venous blood flow (Cochran's $Q = 6.91$, $I^2 = 13\%$, $\text{Tau}^2 = 0.02$, $p = 0.33$) and moderate heterogeneity for arterial measures of blood flow (Cochran's $Q = 27.96$, $I^2 = 50\%$, $\text{Tau}^2 = 0.15$, $p = 0.01$).

A total of ten studies measured peripheral blood flow during a physiological challenge (Fig. 4, Table 1 for details of physiological challenges). A small positive effect was evident for SCG versus controls (SMD = 0.44, 95% CI 0.19, 0.69, $z = 3.40$, $p = 0.0007$). Low heterogeneity amongst

Study	Outcome (area)	SMD [± 95% CI]	Weighting
Venous			
Book et al. [41]	tHb (calf)	0.13 [-0.67, 0.93]	3.4%
Born et al. [72]	tHb (thigh)	0.29 [-0.59, 1.17]	3.1%
Bringard et al. [36]	tHb (calf)	0.55 [-0.27, 1.37]	3.4%
Broatch et al. [34]	tHb (thigh)	0.35 [-0.53, 0.62]	3.0%
Castilho Junior et al. [63]	EF (calf)	0.03 [-0.85, 0.91]	3.1%
Dascombe et al. [70]	tHb (thigh)	0.90 [0.02, 1.79]	3.0%
Dascombe et al. [71]	tHb (forearm)	0.31 [-0.74, 1.37]	2.4%
Ménétrier et al. [37]	EDv (femoral)	0.89 [0.13, 1.64]	3.7%
Oficial-Casado et al. [69]	VBF (popliteal)	-0.60 [-1.50, 0.30]	3.0%
O'Riordan et al. [42]	VBF (femoral)	0.87 [0.25, 1.49]	4.5%
Partsch et al. [64]	EF (calf)	0.12 [-0.54, 0.77]	4.3%
Sear et al. [14]	tHb (thigh)	0.00 [-0.98, 0.98]	2.6%
Venckunas et al. [62]	VER (calf)	0.40 [-0.38, 1.18]	3.6%
Subtotal (95% CI) test for overall effect: z = 3.16 (*p* = 0.002)		**0.37 [0.14, 0.60]**	**43.1%**
Heterogeneity: $Tau^2 = 0.01$; $Chi^2 = 12.43$, df = 12 ($p = 0.41$); $I^2 = 3\%$			
Arterial			
Bochmann et al. [35]	mBF (forearm)	1.76 [0.17, 3.34]	1.2%
Book et al. [41]	ABF (popliteal)	-0.00 [-0.80, 0.80]	3.5%
Broatch et al. [8]	mBF (thigh)	0.29 [-0.35, 0.94]	4.4%
Broatch et al. [34]	MVBF (thigh)	-0.35 [-1.23, 0.54]	3.0%
Broatch et al. [34]	ABF (femoral)	0.11 [-0.77, 0.99]	3.1%
Dorey et al. [65]	MCAv	0.15 [-0.73, 1.03]	3.1%
Kerhervé et al. [40]	mBF (calf)	0.00 [-0.74, 0.74]	3.8%
Mann et al. [61]	ABF (calf)	0.37 [-0.14, 0.88]	5.4%
Ménétrier et al. [37]	ABF (femoral)	2.58 [1.58, 3.58]	2.6%
Morrison et al. [66]	MCAv	-0.04 [-0.75, 0.68]	3.9%
O'Riordan et al. [42]	mBF (calf)	0.44 [-0.16, 1.04]	4.7%
O'Riordan et al. [42]	mBF (thigh)	0.81 [0.19, 1.42]	4.6%
Riexinger et al. [68]	MVBF (calf)	0.00 [-0.69, 0.69]	4.1%
Smale et al. [67]	MCAv	0.20 [-0.51, 0.92]	3.9%
Sperlich et al. [43]	mBF (thigh)	-0.63 [-1.80, 0.54]	2.0%
Venckunas et al. [62]	ABF (calf)	-0.08 [-0.84, 0.69]	3.6%
Subtotal (95% CI) test for overall effect: z = 2.00 (*p* = 0.04)		**0.30 [0.01, 0.59]**	**56.9%**
Heterogeneity: $Tau^2 = 0.19$; $Chi^2 = 34.57$, df = 15 ($p = 0.003$); $I^2 = 57\%$			
Total (95% CI) test for overall effect: z = 3.28 (*p* = 0.001)		**0.32 [0.13, 0.51]**	**100.0%**
Heterogeneity: $Tau^2 = 0.11$; $Chi^2 = 47.28$, df = 28 ($p = 0.01$); $I^2 = 41\%$			

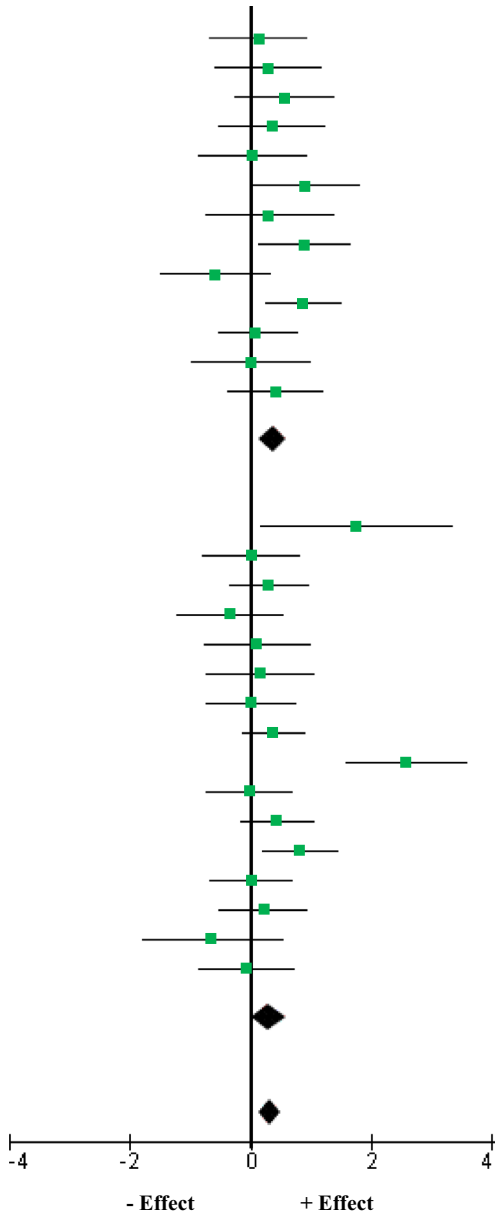

Fig. 2 Forest plot illustrating the overall effects of sports compression garments compared with a control on peripheral blood flow measures in each study. Square *boxes* represent the standardised mean difference (SMD) for each study, with lines representing 95% confidence intervals (CIs). A diamond represents the overall SMD, with its width representing 95% CIs. *ABF* arterial blood flow, *EDv* end diastolic velocity, *EF* ejection fraction, *mBF* muscle blood flow, *MCAv* middle cerebral artery blood flow velocity *MVBF* microvascular blood flow, *tHb* total haemoglobin, *VBF* vein blood flow, *VER* venous emptying rate

studies was observed (Cochran's $Q = 11.42$, $I^2 = 4\%$, $Tau^2 = 0.01$, $p = 0.41$). There were four different measures of peripheral blood flow assessed, with one venous (tHb [14, 34, 41, 70–72]) and three arterial (ABF [34, 41], mBF [8, 35, 40], MCAv [67]) measures of blood flow. For the subgroup analysis, SCG had a small positive effect on venous (SMD = 0.48, 95% CI 0.11, 0.85, $z = 2.51$, $p = 0.01$) and arterial (SMD = 0.44, 95% CI 0.02, 0.86, $z = 2.06$, $p = 0.04$) measures of blood flow. Low heterogeneity amongst studies assessing venous (Cochran's $Q = 3.82$, $I^2 = 0\%$, $Tau^2 = 0.00$, $p = 0.58$) and arterial measures of blood flow (Cochran's $Q = 7.51$, $I^2 = 33\%$, $Tau^2 = 0.09$, $p = 0.190$) was observed.

For peripheral blood flow measures immediately after a physiological challenge, nine studies were analysed (Fig. 5). There was a non-significant trivial difference in peripheral blood flow measures for SCG versus control (SMD = −0.04,

Study	Outcome (area/SCG)	SMD [± 95% CI]	Weighting
Venous			
Book et al. [41]	tHb (calf)	0.80 [-0.04, 1.63]	4.0%
Bringard et al. [36]	tHb (calf)	0.55 [-0.27, 1.37]	4.2%
Oficial-Casado et al. [69]	VBF (popliteal)	-0.65 [-1.55, 0.26]	3.6%
O'Riordan et al. [42]	VBF (femoral/sock)	0.30 [-0.29, 0.90]	5.9%
O'Riordan et al. [42]	VBF (femoral/short)	0.42 [-0.18, 1.02]	5.9%
O'Riordan et al. [42]	VBF (femoral/tight)	0.46 [-0.14, 1.06]	5.9%
Venckunas et al. [62]	VER (calf)	0.00 [-0.77, 0.77]	4.5%
Subtotal (95% CI) test for overall effect: z = 2.12 (*p* = 0.03)		**0.31 [0.02, 0.60]**	**34.0%**
Heterogeneity: $Tau^2 = 0.02$; $Chi^2 = 6.91$, df = 6 ($p = 0.33$); $I^2 = 13\%$			
Arterial			
Bochmann et al. [35]	mBF (forearm)	1.52 [0.01, 3.03]	1.6%
Book et al. [41]	ABF (popliteal)	-0.21 [-1.01, 0.59]	4.3%
Broatch et al. [8]	mBF (thigh)	0.50 [-0.15, 1.14]	5.4%
Broatch et al. [34]	MVBF (thigh)	-0.30 [-1.18, 0.58]	3.8%
Broatch et al. [34]	ABF (femoral)	0.02 [-0.86, 0.89]	3.8%
Dorey et al. [65]	MCAv	-0.12 [-1.00, 0.76]	3.8%
Kerhervé et al. [40]	mBF (calf)	0.53 [-0.23, 1.28]	4.6%
Morrison et al. [66]	MCAv	-0.08 [-0.80, 0.63]	4.9%
O'Riordan et al. [42]	mBF (calf/sock)	0.44 [-0.16, 1.04]	5.9%
O'Riordan et al. [42]	mBF (thigh/short)	0.81 [0.19, 1.42]	5.7%
O'Riordan et al. [42]	mBF (thigh/tight)	0.59 [-0.02, 1.19]	5.8%
Riexinger et al. [68]	MVBF (calf)	-0.31 [-1.01, 0.39]	5.0%
Smale et al. [67]	MCAv	0.11 [-0.61, 0.83]	4.9%
Sperlich et al. [43]	mBF (thigh)	-0.53 [-1.69, 0.63]	2.5%
Venckunas et al. [62]	ABF (calf)	-1.20 [-2.05, -0.36]	4.0%
Subtotal (95% CI) test for overall effect: z = 0.83 (*p* = 0.41)		**0.12 [-0.16, 0.40]**	**66.0%**
Heterogeneity: $Tau^2 = 0.15$; $Chi^2 = 27.96$, df = 14 ($p = 0.01$); $I^2 = 50\%$			
Total (95% CI) test for overall effect: z = 1.73 (*p* = 0.08)		**0.18 [-0.02, 0.39]**	**100%**
Heterogeneity: $Tau^2 = 0.10$; $Chi^2 = 35.70$, df = 21 ($p = 0.02$); $I^2 = 41\%$			

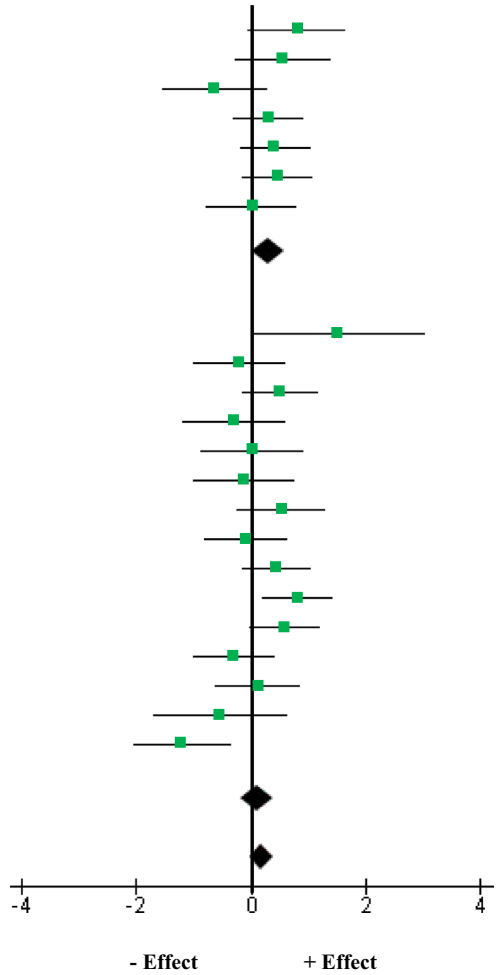

Fig. 3 Forest plot illustrating the effects of sports compression garments compared with a control on peripheral blood flow measures at rest prior to a physiological challenge. Square boxes represent the standardised mean difference (SMD) for each study, with lines representing 95% confidence intervals (CIs). A diamond represents the overall SMD, with its width representing 95% CIs. *ABF* arterial blood flow, *mBF* muscle blood flow, *MCAv* middle cerebral artery blood flow velocity, *MVBF* microvascular blood flow, *SCG* sports compression garment type, *tHb* total haemoglobin, *VBF* vein blood flow, *VER* venous emptying rate

95% CI − 0.43, 0.34, $z = 0.23$, $p = 0.82$) immediately after a physiological challenge. Moderate heterogeneity amongst studies was observed (Cochran's $Q = 17.77$, $I^2 = 55\%$, $Tau^2 = 0.19$, $p = 0.02$). A total of six different measures of peripheral blood flow were assessed, with two venous (EF [63, 64], VBF [69]) and four arterial (MVBF [34, 68], MCAv [65, 66], mBF [40], ABF [61]). For the subgroup analysis, there were non significant small and trivial differences between SCG and control groups for venous (SMD = − 0.43, 95% CI − 1.32, 0.47, $z = 0.94$, $p = 0.35$) and arterial (SMD = 0.12, 95% CI − 0.26, 0.51, $z = 0.63$, $p = 0.53$) measures of blood flow, respectively. Moderate heterogeneity amongst studies assessing venous (Cochran's $Q = 6.04$, $I^2 = 67\%$, $Tau^2 = 0.42$, $p = 0.05$) and arterial measures of blood flow (Cochran's $Q = 8.55$, $I^2 = 42\%$, $Tau^2 = 0.10$, $p = 0.13$) was observed.

A total of six studies measured peripheral blood flow during recovery from a physiological challenge (Fig. 6). There was a non-significant small effect for SCG versus control (SMD = 0.25, 95% CI − 0.45, 0.95, $z = 0.69$, $p = 0.49$). High heterogeneity amongst studies was observed (Cochran's $Q = 68.85$, $I^2 = 85\%$, $Tau^2 = 1.18$, $p < 0.00001$). There were six different measures of peripheral blood flow including three venous (tHb [34, 41], EDv [37], VER [62]) and three arterial (MVBF [34, 68], ABF [34, 37, 62, 41], mBF [43]). For the subgroup analysis, SCG had a medium positive effect on venous blood flow (SMD = 0.67, 95% CI 0.17, 1.17, $z = 2.61$, $p = 0.009$). In contrast, there was a non-significant trivial difference in arterial blood flow between

Study	Challenge	Time/ Intensity	Outcome (area)	SMD [± 95% CI]	Weighting
Venous					
Book et al. [41]	Plantar Flexion	5 min	tHb (calf)	0.48 [-0.34, 1.29]	9.2%
Born et al. [72]	Ice Skating	Last min	tHb (thigh)	0.29 [-0.59, 1.17]	7.9%
Broatch et al. [34]	Cycling	Set 1	tHb (thigh)	0.49 [-0.41, 1.38]	7.7%
Dascombe et al. [70]	Running	18 km/h	tHb (thigh)	1.25 [0.32, 2.18]	7.1%
Dascombe et al. [71]	Kayaking	4 min PT	tHb (forearm)	0.31 [-0.74, 1.37]	5.6%
Sear et al. [14]	Running	31 – 45 min	tHb (thigh)	0.00 [-0.98, 0.98]	6.4%
Subtotal (95% CI) test for overall effect: z = 2.51 (*p* = 0.01)				**0.48 [0.11, 0.85]**	**43.9%**
Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 3.82$, df = 5 ($p = 0.58$); $I^2 = 0\%$					
Arterial					
Bochmann et al. [35]	Handgrip	65 min	mBF (forearm)	2.64 [0.70, 4.57]	1.7%
Book et al. [41]	Plantar Flexion	5 min	ABF (popliteal)	0.11 [-0.69, 0.91]	9.5%
Broatch et al. [8]	Cycling	Set 4	mBF (thigh)	0.36 [-0.26, 0.99]	15.2%
Broatch et al. [34]	Cycling	Set 2	ABF (femoral)	0.75 [-0.16, 1.66]	7.4%
Kerhervé et al. [40]	Running	10 km	mBF (calf)	0.00 [-0.74, 0.74]	11.0%
Smale et al. [67]	Cycling	70%	MCAv	0.55 [-0.18, 1.28]	11.3%
Subtotal (95% CI) test for overall effect: z = 2.06 (*p* = 0.04)				**0.44 [0.02, 0.86]**	**56.1%**
Heterogeneity: $Tau^2 = 0.09$; $Chi^2 = 7.51$, df = 5 ($p = 0.19$); $I^2 = 33\%$					
Total (95% CI) test for overall effect: z = 3.40 (*p* = 0.0007)				**0.44 [0.19, 0.69]**	**100.0%**
Heterogeneity: $Tau^2 = 0.01$; $Chi^2 = 11.42$, df = 11 ($p = 0.41$); $I^2 = 4\%$					

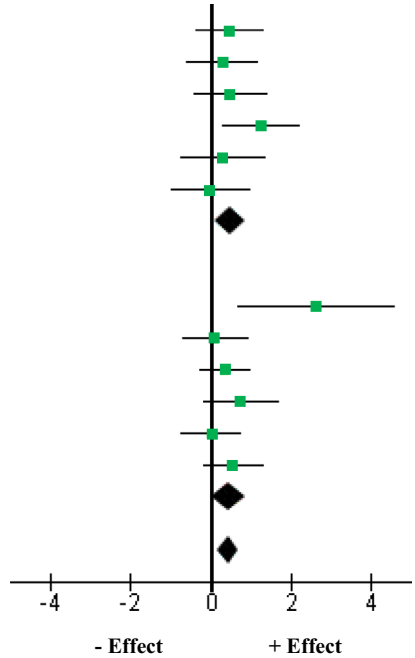

Fig. 4 Forest plot illustrating the effects of sports compression garments compared with a control on peripheral blood flow measures during a physiological challenge. Square boxes represent the standardised mean difference (SMD) for each study, with lines representing 95% confidence intervals (CIs). A diamond represents the overall SMD, with its width representing 95% CIs. *ABF* arterial blood flow, *mBF* muscle blood flow, *MCAv* middle cerebral artery blood flow velocity, *tHb* total haemoglobin

Study	Challenge	Outcome	SMD [± 95% CI]	Weighting
Venous				
Castilho Junior et al. [63]	Running	EF (calf)	0.02 [-0.85, 0.90]	10.0%
Oficial-Casado et al. [69]	Running	VBF (popliteal)	-1.45 [-2.46, -0.44]	8.6%
Partsch et al. [64]	Walking	EF (calf)	0.02 [-0.78, 0.82]	10.9%
Subtotal (95% CI) test for overall effect: z = 0.94 (*p* = 0.35)			**-0.43 [-1.32, 0.47]**	**29.5%**
Heterogeneity: $Tau^2 = 0.42$; $Chi^2 = 6.04$, df = 2 ($p = 0.05$); $I^2 = 67\%$				
Arterial				
Broatch et al. [34]	Cycling	MVBF (thigh)	-0.87 [-1.80, 0.05]	9.4%
Dorey et al. [65]	HUT Post Cycle	MCAv	0.49 [-0.40, 1.38]	9.8%
Kerhervé et al. [40]	Running	mBF (calf)	-0.10 [-0.84, 0.64]	11.7%
Mann et al. [61]	Running	ABF (calf)	0.57 [0.06, 1.09]	15.1%
Morrison et al. [66]	Supine to Stand	MCAv	0.00 [-0.72, 0.72]	12.1%
Riexinger et al. [68]	Plyo + Ecc	MVBF (calf)	0.28 [-0.41, 0.98]	12.1%
Subtotal (95% CI) test for overall effect: z = 0.63 (*p* = 0.53)			**0.12 [-0.26, 0.51]**	**70.5%**
Heterogeneity: $Tau^2 = 0.10$; $Chi^2 = 8.55$, df = 5 ($p = 0.13$); $I^2 = 42\%$				
Total (95% CI) test for overall effect: z = 0.23 (*p* = 0.82)			**-0.04 [-0.43, 0.34]**	**100.0%**
Heterogeneity: $Tau^2 = 0.19$; $Chi^2 = 17.77$, df = 8 ($p = 0.02$); $I^2 = 55\%$				

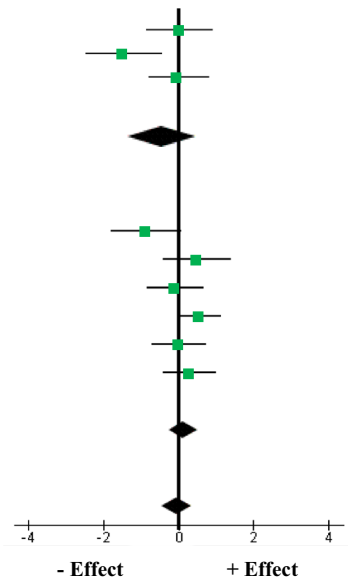

Fig. 5 Forest plot illustrating the effects of sports compression garments compared with a control on peripheral blood flow measures immediately after a physiological challenge. Square boxes represent the standardised mean difference (SMD) for each study, with lines representing 95% confidence intervals (CIs). A diamond represents the overall SMD, with its width representing 95% CIs. *ABF* arterial blood flow, *EF* ejection fraction, *mBF* muscle blood flow, *MCAv* middle cerebral artery blood flow velocity, *MVBF* microvascular blood flow, *Plyo + Ecc* plyometric and eccentric exercise, *VBF* vein blood flow

Study	Challenge	Time	Outcome (area)	SMD [± 95% CI]	Weighting
Venous					
Book et al. [41]	Plantar Flexion	5 min	tHb (calf)	0.26 [-0.54, 1.06]	9.5%
Broatch et al. [34]	Cycling	60 min	tHb (thigh)	0.19 [-0.69, 1.07]	9.3%
Ménétrier et al. [37]	Cycling	9 min	EDv (femoral)	1.22 [0.43, 2.01]	9.6%
Venckunas et al. [62]	Running	30 min	VER (calf)	0.94 [0.12, 1.75]	9.5%
Subtotal (95% CI) test for overall effect: z = 2.62 (*p* = 0.009)				**0.67 [0.17, 1.17]**	**37.8%**
Heterogeneity: $Tau^2 = 0.08$ $Chi^2 = 4.43$, df = 3 ($p = 0.22$); $I^2 = 32\%$					
Arterial					
Book et al. [41]	Plantar Flexion	5 min	ABF (popliteal)	-0.12 [-0.93, 0.68]	9.5%
Broatch et al. [34]	Cycling	60 min	MVBF (thigh)	-1.88 [-2.97, -0.79]	8.6%
Broatch et al. [34]	Cycling	60 min	ABF (femoral)	0.07 [-0.80, 0.95]	9.3%
Ménétrier et al. [37]	Cycling	9 min	ABF (femoral)	4.32 [2.94, 5.69]	7.7%
Riexinger et al. [68]	Plyo + Ecc	30 min	MVBF (calf)	-0.44 [-1.14, 0.27]	9.8%
Sperlich et al. [43]	Cycling	10 min	mBF (thigh)	-1.64 [-3.03, -0.25]	7.6%
Venckunas et al. [62]	Running	30 min	ABF (calf)	0.04 [-0.73, 0.81]	9.6%
Subtotal (95% CI) test for overall effect: z = 0.03 (*p* = 0.98)				**0.02 [-1.06, 1.09]**	**62.2%**
Heterogeneity: $Tau^2 = 1.85$; $Chi^2 = 55.86$, df = 6 ($p < 0.00001$); $I^2 = 89\%$					
Total (95% CI) test for overall effect: z = 0.69 (*p* = 0.49)				**0.25 [-0.45, 0.95]**	**100.0%**
Heterogeneity: $Tau^2 = 1.18$; $Chi^2 = 68.85$, df = 10 ($p < 0.00001$); $I^2 = 85\%$					

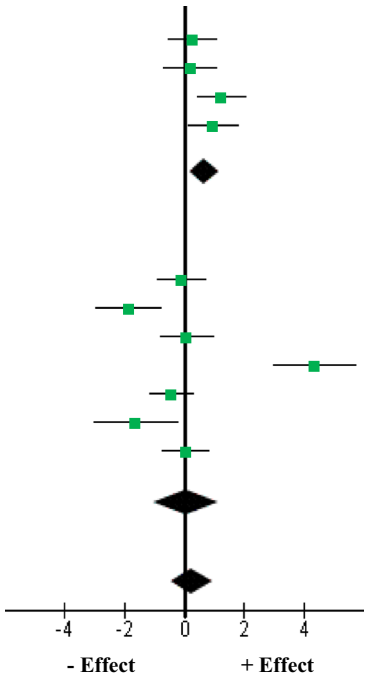

Fig. 6 Forest plot illustrating the effects of sport compression garments compared with a control on peripheral blood flow measures in recovery. Square boxes represent the standardised mean difference (SMD) for each study, with lines representing 95% confidence intervals (CIs). A diamond represents the overall SMD, with its width representing 95% CIs. *ABF* arterial blood flow, *EDv* end diastolic velocity, *mBF* muscle blood flow, *MVBF* microvascular blood flow, *Plyo + Ecc* plyometric and eccentric exercise, *tHb* total haemoglobin, *VER* venous emptying rate

SCG and control groups (SMD = 0.02, 95% CI − 1.06, 1.09, $z = 0.03$, $p = 0.98$). Moderate and high heterogeneity was observed for studies assessing venous (Cochran's $Q = 4.43$, $I^2 = 32\%$, $Tau^2 = 0.08$, $p = 0.22$) and arterial (Cochran's $Q = 55.64$, $I^2 = 89\%$, $Tau^2 = 1.85$, $p < 0.00001$) measures of blood flow, respectively.

4 Discussion

The main findings from this systematic review with a meta-analysis were that SCG enhanced measures of peripheral blood flow, with changes evident at rest, during and in recovery from a physiological challenge. The use of SCG during a physiological challenge had positive effects on measures of venous and arterial blood flow. Although SCG did not improve overall peripheral blood flow at rest and in recovery, there was an effect for measures of venous blood flow. Additionally, no effects of SCG were observed on venous or arterial blood flow immediately after a physiological challenge.

4.1 At Rest

Regarding changes in peripheral blood flow at rest, our results showed no overall effect of SCG to alter peripheral blood flow (Fig. 3). However, there was a positive effect for venous measures of blood flow. The external compression applied to the limb with SCG is reported to decrease the vein diameter, which in turn will improve valve competence and reduce venous reflux [31]. As a result, blood is diverted from superficial veins, through perforator veins and into deep veins, leading to an increase in venous velocity, reduced venous pooling and improved venous flow. Our analysis supports these mechanisms at rest, when there is minimal lower-limb muscle contraction (i.e. little muscle pump activation), showing that SCG appear to augment venous blood flow and muscle pump functionality as is evident from positive changes in tHb [36, 41] and femoral VBF [42] (Fig. 3).

At rest, skeletal muscle requires approximately 20% of overall cardiac output, with most blood flow directed to the brain, kidney and liver [44]. Further, arterial blood flow is tightly coupled to the metabolic demand of the muscles [73]. Thus, the magnitude of potential changes in arterial blood flow with compression at rest might be difficult to detect because of the lack of overall blood flow distribution to the

compressed areas (i.e. the limbs). Additionally, resting blood flow is influenced by several factors, including time of day [74, 75], environmental conditions [76–78], nutrient intake [79–81], posture [82, 83] and the rest time before data collection [84, 85]. Therefore, to quantify the SCG effect on resting blood flow, it is vital to control these factors. Some studies failed to address these potential confounders, which may help explain the variability in resting arterial blood flow in the analysed studies (Fig. 3). For example, 50% of the studies analysed did not provide information on the duration of fasting before collecting resting blood flow measures [8, 35, 36, 40, 67–69]. Research has shown that the composition of a meal (i.e. mixed nutrient, high glucose, high fat) can either increase [80, 86] or decrease [79, 81, 87] measures of arterial blood flow. Further, in the studies that did provide fasting durations, they were between 2 [62, 66] and 3 h [34, 43, 65] before measuring blood flow. These times might not be sufficient as alterations in blood flow can be evident for up to 4 h postprandial [81, 87–89]. In support of the current findings, a recent meta-analysis concluded that compression garments do not alter resting hemodynamic variables, including heart rate, cardiac output, blood pressure and systemic vascular resistance [50]. The authors proposed that the lack of disturbance to central haemodynamics at rest (i.e. low metabolic demand of skeletal muscle) may explain these findings. Additionally, they suggest the efficacy of compression is dependent on the degree of physiological challenge and alterations to central hemodynamics. These proposed mechanisms are relevant to and help explain the lack of compression-induced changes in arterial measures of blood flow identified at rest in this review (Fig. 3).

4.2 During Exercise

The results from our review indicate that SCG enhance peripheral blood flow during a physiological challenge. During exercise, skeletal muscle contraction compresses the intramuscular veins and effectively enhances venous flow [90, 91]. In turn, the maintenance of venous return is due to the increases in intramuscular pressure from rhythmic muscle contraction (skeletal muscle pump) [91]. The application of external compression (i.e. SCG) may further augment the skeletal muscle pump and subsequently increase venous flow. Compression-induced increases in venous return could enhance stroke volume and cardiac output via increased end-diastolic filling [51]. Our findings of enhanced venous flow during a physiological challenge, combined with previous reports of increased cardiac output [92–94] and stroke volume [93–95] with compression, support this mechanism. As stroke volume is a key factor underlying endurance performance [96], the augmentation of venous blood flow during exercise with compression may serve to improve exercise performance. However, this mechanism warrants further investigation as contradictory findings have been reported for changes in stroke volume [41, 94] and endurance performance [97, 98] with compression. Furthermore, enhanced venous blood flow with compression is suggested to assist in clearing circulating metabolites (e.g. lactate) from the skeletal muscle during exercise [99], thus increasing exercise capacity at higher intensities.

Our subgroup analysis also highlights enhanced arterial blood flow during a physiological challenge with SCG (Fig. 4). Exercise is associated with a significant rise in skeletal muscle blood flow, by almost 100-fold [100], which is necessary to match the metabolic demand for oxygen in the contracting muscles [44, 73]. Additionally, the energy requirements of skeletal muscle are fueled by blood-borne nutrients (i.e. blood glucose, free fatty acids), with increased muscle blood flow essential for their supply during exercise [101, 102]. Therefore, compression-induced increases in arterial and muscle blood flow may enhance exercise performance via improved oxygen and nutrient delivery to the muscle. For example, Sear et al. [14] reported increased muscle oxygenation during intermittent high-intensity running when wearing SCG, resulting in an 8.5% improvement in total distance covered. However, research demonstrating an improvement in nutrient delivery to the muscle during exercise with SCG is lacking and requires further investigation. Compression may also provide an ergogenic benefit on cerebral blood flow, as measured by MCAv [67]. The data from Smale et al. [67] showed small beneficial effects of SCG on MCAv at higher exercise intensities (70% maximal aerobic power; Fig. 4) in conjunction with enhanced cognitive performance, but not for cycling time-trial performance. Alterations in mean arterial pressure, influenced by changes in cardiac output and total peripheral resistance [103], are an important stimulus to elicit changes in cerebral blood flow [104]. Although Smale et al. [67] observed no changes in mean arterial pressure with SCG, previous work has highlighted positive changes in cardiac output [92, 94] and total peripheral resistance [37] with compression garments. Thus, changes in mean arterial pressure due to compression might contribute to the observed benefits in MCAv with SCG. However, as findings are limited to this single study [67], more research is warranted.

It is also important to acknowledge the constraints in achieving accurate measures of peripheral blood flow during exercise. Although NIRS allows for the continuous assessment of tHb and has been validated against other techniques assessing blood flow [105, 106], it is limited in the information it can provide on peripheral blood velocity, flow and vessel structure. However, a limitation of more robust methods to assess alterations in peripheral blood flow, such as Doppler ultrasound or venous occlusion plethysmography, is that participants are required to stop exercising to obtain accurate measurements. Nonetheless, despite the

methodological differences between studies (i.e. exercise intervention, garment style, measurement of blood flow), there was an overall effect of SCG to improve venous and arterial blood flow during a physiological challenge.

4.3 Immediately Post

The subgroup analysis for peripheral blood flow at the immediately post timepoint showed that SCG had a trivial effect on venous or arterial measures of blood flow. The lack of an effect of SCG on peripheral blood flow immediately post a physiological challenge is inconsistent with the improvements reported during the subsequent recovery period (Fig. 6). In addition, this finding contrasts with those of a recent meta-analysis that concluded central haemodynamic measures (e.g. stroke volume, heart rate) after a physiological challenge are improved with compression garments because of enhanced venous flow [50]. These contradictory findings likely stem from the combination of only three studies analysed for venous blood flow, the observed large negative effects of SCG on venous flow post-running [69], and the variability in blood flow measurement techniques (i.e. magnetic resonance imaging [MRI], CEU, NIRS, air plethysmography) and measures of arterial blood flow (e.g. macrovascular and microvascular). Still, caution is advised in interpreting the findings from Oficial-Casado et al. [69]. Specifically, the pressure applied was not measured but only reported as medium-pressure or high-pressure garments. The pressure exerted in these garments may have reduced venous blood flow by applying excess pressure. In support of this, a greater reduction in venous blood flow was evident with the high-pressure garments [69]. Although speculative, it highlights the importance of SCG research to measure the degree of pressure, or, at the very minimum, report the manufacturer values. Furthermore, the intensity/mode of exercise (i.e. walking [64] vs running [63, 69]) may explain the lack of effect observed for venous blood flow as compression-induced changes in blood flow are likely influenced by the degree of the physiological challenge [50]. Additionally, SCG have been reported to have divergent effects on macrovascular (arterial) and microvascular (capillaries) blood flow [34]. Although compression increases large artery blood flow, likely via a myogenic response, capillaries are more susceptible to compression because of their lack of vascular smooth muscle and connective tissue [34]; thus, reducing microvascular blood flow, volume and velocity [34]. Our observations of no change in arterial blood flow immediately post timepoint might be explained by these divergent effects (i.e. increased arterial blood flow and reduced microvascular blood flow) (Fig. 5); however, further study is needed to understand the effect of SCG on microvascular measures.

4.4 Recovery

The findings from this meta-analysis support the use of SCG to enhance venous blood flow in recovery, but not arterial blood flow. The aim of recovery techniques following a physiological challenge, such as exercise, is to return the body to homeostasis [107]. A frequently proposed mechanism attributed to restoring indices of exercise recovery (i.e. exercise performance, muscle soreness) is increased blood flow [108–110]. Augmented venous flow is hypothesised to facilitate the removal of metabolic by-products from the previously active muscles [108], which might otherwise contribute to the formation of exercise-induced oedema [111]. As we have shown SCG to augment venous flow in recovery, this mechanism is further supported by lower levels of metabolic by-products including creatine kinase [23, 112, 113] and lactate dehydrogenase [23, 25] when SCG are worn post-exercise. Compression-induced increases in venous return may also serve as a protective mechanism against post-exercise hypotension, which can persist for several hours [114] and if individuals remain in a supine position (i.e. recovery after exercise) [93]. Post-exercise hypotension, as observed in trained [115, 116] and untrained [117, 118] individuals, is characterised by a reduction in blood pressure and occurs as a result of a combination of an inactive skeletal muscle pump [119], pooling of blood in previously active muscles [116], decreased end-diastolic filling [120] and reduced stroke volume [116]. The benefit of SCG on venous flow described in this review, combined with reports of improved stroke volume [50, 93, 94] and blood pressure [93, 121] with compression garments, highlights the beneficial effects SCG may have in preventing post-exercise hypotension.

Our analysis shows no beneficial effects of SCG on arterial and muscle blood flow during recovery (Fig. 6). Similar to the time point immediately post, the lack of effect during recovery likely stems from the divergent effects of SCG on macro and microvascular measures of blood flow. However, caution is advised in interpreting the lack of change in arterial/muscle blood flow as there was a high level of heterogeneity between the studies ($I^2 = 89\%$; Fig. 6). Nonetheless, an enhanced muscle blood flow may improve oxygen and nutrient delivery in recovery, which could aid phosphocreatine [122] and glycogen resynthesis [123]. However, this response is yet to be supported in compression garment research as studies investigating muscle oxygenation with SCG in recovery are limited, with contradictory findings [62, 124]. Additionally, only one study has investigated the effect of SCG on nutrient delivery during recovery, with no change in muscle glucose uptake observed with compression [43]. This finding also coincided with reduced muscle blood flow [43] with compression (Fig. 6). The high degree of pressure exerted by the sports compression shorts (37 mmHg) [43],

higher than reported in other compression garment studies [8, 62, 67, 70, 72], likely mechanically hindered muscle blood flow. In support, microvascular blood flow is attenuated at 30 and 60 min post-exercise with SCG (Fig. 6) [34, 68]. Considering the microvascular network (i.e. capillaries) is responsible for nutrient and oxygen exchange in the muscle, a compression-induced reduction in microvascular blood flow may have implications for nutrient delivery, and long-term muscle adaptations reliant on adequate muscle perfusion (e.g. muscle protein synthesis) [34, 125]. Our review highlights that no research has investigated the effect of SCG on measures of peripheral blood flow beyond 60 min of use after a physiological challenge. Considering the inflammatory response initiates in the early hours (1–4 h) [126] following exercise, further assessment of compression-induced changes in blood flow during this time is warranted. In turn, this will provide valuable insight into the proposed mechanism (i.e. increased blood flow) attributed to the effectiveness of SCG as a post-exercise recovery strategy.

4.5 Limitations

Although this meta-analysis highlighted changes in peripheral blood flow with SCG, limitations of this review must be considered when interpreting these findings. First, the overall quality of the studies included was moderate (5.8 arbitrary units). Another major issue with compression garment research is the lack of blinding of participants and researchers. The difficulty here lies in the high degree of pressure exerted by SCG, as the participant can quickly identify if receiving the SCG intervention. This is highlighted in the current review as none of the 22 studies incorporated blind allocation, blinding of subjects or blinding of the researchers who administered the intervention. With the inadequate blinding in compression research, the placebo effect for participants and an increased risk of bias and interpretation by researchers cannot be discounted. Similar concerns have been highlighted in previous SCG reviews [1, 26, 27, 46, 49, 127, 128]. As placebo interventions may alter cardiovascular function (i.e. the tone of blood vessels, blood pressure, heart rate variability) [129, 130], future compression research should incorporate a placebo intervention (e.g. sham drink with reported benefits similar to SCG) and the blinding of researchers when analysing data (i.e. unaware if data are from placebo, control or SCG intervention). These recommendations would serve to improve the quality research assessing the potential benefits of effects of SCG.

The high heterogeneity observed in this review might be explained by methodological variations, particularly in the SCG details. For example, the garment types included upper-body and lower-body garments, with lower-body garments divided even further into compression socks, tights and shorts. We have previously shown that sports compression tights were the most effective garment type, compared with sports compression socks and sports compression shorts, to improve resting lower-limb blood flow [42]. In addition, some participants may not have received adequate pressure, with only 14 studies measuring exact pressures applied (Table 1). Other studies reported pressure values indicated by the garment manufacturer [37, 63, 65, 68] or failed to report any pressure values [61, 69, 71]. This issue is problematic as differences in limb dimensions may lead to wide inter-individual variations in pressure and could result in individuals receiving insufficient or excessive levels of pressure [2]. In addition, manufacturer sizing guidelines are typically acquired from crude anthropometric measurements (i.e. height, mass, limb circumference), and these can result in large pressure ranges exerted by a single garment size [2]. The level of pressure exerted by SCG is a crucial consideration and should be measured in future studies as pressure and gradient distributions are critical for the garments' effectiveness [28, 131].

Another limitation to consider is the different physiological challenges in this review, which included an orthostatic test [66], rhythmic movement exercises [35, 41], walking [64], running [40, 61–63, 69, 70], cycling [8, 34, 37, 65], kayaking [71], ice skating [72] and plyometric/eccentric exercise [68]. Considering the efficacy of SCG is suggested to be dependent on the degree of physiological challenge and alterations to haemodynamics [50], the type, duration and intensity of a physiological challenge may influence the observed effects of compression-induced changes in measures of peripheral blood flow. For example, there appears to be a greater benefit of SCG to improve tHb measures as exercise intensity [70] or duration/set number [8] increases. However, this is not evident in other studies for the same measure of tHb [14, 71, 72] or arterial measures of blood flow [8, 34, 35, 67]. The unclear relationship suggests that other factors (i.e. garment pressure, garment type) may contribute more to the effectiveness of SCG to alter peripheral blood flow than the the physiological challenge intensity or duration. The assessment of peripheral blood flow also varied with different techniques used, including NIRS [8, 34, 36, 40–42, 70–72], Doppler ultrasound [34, 37, 41, 42, 65–67], plethysmography [35, 61–64], MRI [69], CEU [34], intravoxel incoherent motion [68] and positron emission tomography [43]. Considering the divergent effects of SCG on macrovascular and microvascular blood flow [34], and the limitation of techniques in assessing changes in vessel structure (i.e. NIRS, plethysmography), future research should aim to use more accurate techniques (e.g. CEU, Doppler ultrasound, MRI) to quantify compression-induced changes in peripheral blood flow. Finally, the range of participants described in the included studies varied from healthy individuals [35, 41, 65, 66, 68] to recreationally trained/physically active [8, 34, 43, 61–63] and to elite/trained athletes

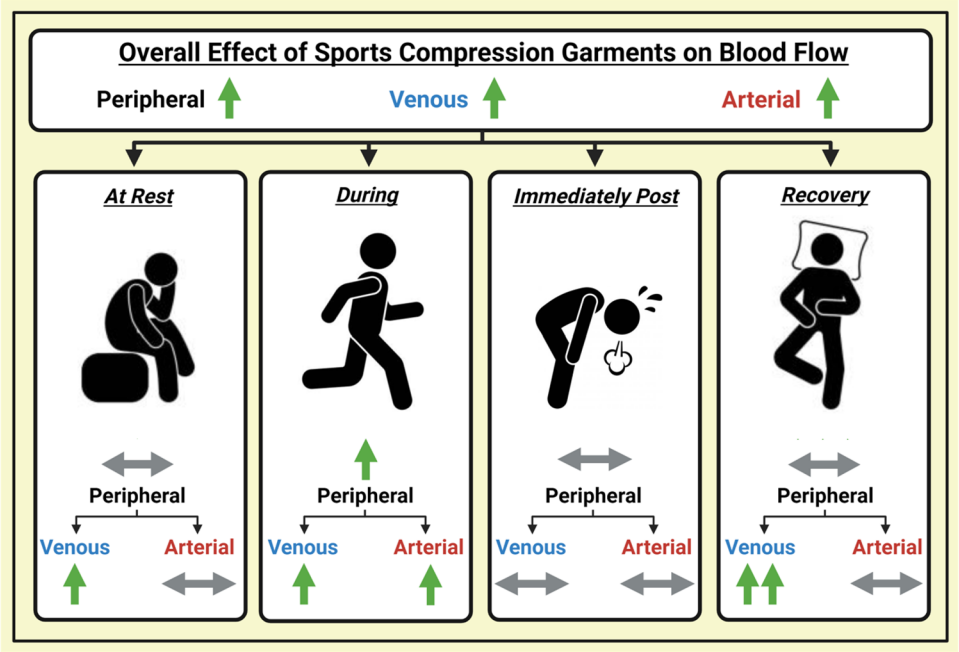

Fig. 7 Summary of the effects of sports compression garments on peripheral blood flow. Created with Biorender.com. Neutral arrow indicates no effect, one up arrow indicates a small positive effect, two up arrows indicate a medium positive effect

[36, 37, 40, 42, 67, 69, 70, 72]. As differences in peripheral blood flow exist between trained and untrained participants [132–135], some individuals might be more susceptible to compression-induced blood flow changes. However, research with compression garments is required to support this theory.

5 Conclusions

Sports compression garments appear to enhance measures of venous blood flow at rest, during (plus arterial measures) and in recovery from a physiological challenge (Fig. 7). In addition, at immediately after a physiological challenge, compression garments did not alter peripheral blood flow measures. The findings from this review support the mechanism of compression-induced changes in peripheral blood flow and highlight that SCG are most beneficial when worn during a physiological challenge. However, additional high-quality research that addresses the limitations of compression garment research highlighted in this review is warranted. These limitations include measuring the pressure levels of the garments, blinding of participants to limit the placebo effect, blinding of researchers to limit the risk of bias, delineating between compression-induced changes in macrovascular and microvascular blood flow and utilising more accurate techniques to assess peripheral blood flow (e.g. CEU, Doppler ultrasound, MRI).

Declarations

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Consent to participate Not applicable.

Consent for publication Not applicable.

Availability of data and material All data analysed and reported in this systematic review and meta-analysis are available from the corresponding author on reasonable request.

Code availability Not applicable.

Author contributions SFO, JRB, SLH and DJB conceptualised the review and criteria. SFO and JRB completed the screening, data extraction and data analysis and drafted the manuscript. All authors reviewed, refined and approved the final manuscript.

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